Supplementary Material

Interpretable Multimodal Analysis of Lexical Tone–Chord Alignment in Vietnamese Music

This supplement gives the full model output behind the main paper. All models use the 583 listing-only chord sheets (symbolic analyses) or the 88 aligned recordings (audio analyses) described there. GEE models use a logit link, exchangeable working correlation and robust sandwich standard errors, clustered by song.

S1. Full coefficients of the extended GEE model (Eq. 4)

Outcome: syllable carries a level tone. Change and bolero are 0/1 indicators (reference: V-pop, no chord change). Continuous controls are as defined in the paper.

Table S1: Extended GEE model: log-odds coefficients.

Term	β	SE	p	e^{β}
Intercept	0.650	0.045	5.2×10^{-47}	1.92
Chord change	0.045	0.031	0.15	1.05
Bolero	0.130	0.023	1.3×10^{-8}	1.14
Chord change × bolero	0.295	0.047	5.4×10^{-10}	1.34
Line-final syllable	0.958	0.040	3.0×10^{-125}	2.61
Line-initial syllable	-0.265	0.034	1.3×10^{-14}	0.77
Relative position in line	-0.476	0.036	6.6×10^{-41}	0.62
Line length (syllables)	0.006	0.002	0.020	1.01
log chord-segment length	0.009	0.018	0.62	1.01
Minor key	0.049	0.025	0.049	1.05
Relative line position in song	-0.050	0.023	0.027	0.95
Song-final line	0.055	0.028	0.053	1.06

Derived odds ratios of a level tone at a chord change: V-pop $e^{\beta_1} = 1.05$ [0.98, 1.11]; bolero $e^{\beta_1+\beta_3} = 1.40$ [1.30, 1.51]; ratio $e^{\beta_3} = 1.34$ [1.22, 1.47].

S2. Six-tone and register models

Line-internal GEE of chord change on tone representation, with relative position and line length as controls, fitted per genre. Lower QIC is better.

Table S2: Model comparison by QIC.

Tone representation	Bolero	V-pop
Poetic class (level/sharp)	37,267	47,873
Register (high/low)	37,447	47,360
Class + register	37,265	47,361
Six tones	37,237	47,298

Table S3: Six-tone odds ratios of a chord change (reference: *ngang*); p raw and Holm-adjusted significance over the ten tests.

	Bolero			V-pop		
Tone	OR	p	Holm	OR	p	Holm
huyền	1.00	0.99	no	0.62	1.5×10^{-23}	yes
sắc	0.77	3.6×10^{-6}	yes	1.16	0.001	yes
hỏi	0.52	1.9×10^{-8}	yes	0.53	3.6×10^{-17}	yes
ngã	0.56	3.3×10^{-10}	yes	0.78	3.2×10^{-5}	yes
nặng	0.63	5.2×10^{-10}	yes	0.51	2.4×10^{-28}	yes

Wald tests. Bolero: *huyền* = *ngang* $p = 0.99$; four sharp tones equal $p = 1.2 \times 10^{-5}$; all within-class equalities $\chi^2_4 = 26.0$, $p = 3.2 \times 10^{-5}$. V-pop: $\chi^2_4 = 249.0$, $p = 1.1 \times 10^{-52}$. Level-minus-sharp log-odds: bolero 0.485 ($p = 5.8 \times 10^{-20}$), V-pop 0.113 ($p = 0.004$). In the class + register model, V-pop high register OR 1.77, level class OR 1.00; bolero high register OR 1.05, level class OR 1.47.

S3. *Lục bát* line types

Extended GEE with a change × genre × couplet-line interaction (couplet line: a 6- or 8-syllable line in a 6–8 pair). Couplet lines hold 2.8% of bolero and 2.3% of V-pop syllables.

Table S4: Chord-change odds ratios by line type.

Quantity	OR [95% CI]	p
Bolero change, couplet lines	1.45 [1.16, 1.82]	0.001
Bolero change, other lines	1.40 [1.30, 1.51]	3.0×10^{-18}
V-pop change, couplet lines	1.22 [0.93, 1.60]	0.14
V-pop change, other lines	1.04 [0.98, 1.11]	0.19
Change × bolero, couplet lines	1.19 [0.84, 1.69]	0.34
Change × bolero, other lines	1.35 [1.23, 1.48]	5.8×10^{-10}
Change × bolero × couplet	0.88 [0.62, 1.26]	0.49

Song-level *lục bát* conformity (standardised; 50 songs with enough couplets): change × conformity $\beta = 0.059$ ($p = 0.52$); change × bolero × conformity $\beta = -0.013$ ($p = 0.91$).

S4. Genre classification: PhoBERT plus each feature group

Logistic regression (balanced class weights; $C = 0.05$ with PhoBERT), standardisation fitted within training folds, 5-fold stratified group CV, mean ROC AUC over 5 seeds; gain over PhoBERT with 95% bootstrap CI over songs.

Table S5: ROC AUC and gain over PhoBERT.

	Composer groups		Strict groups	
Model	AUC	Gain [95% CI]	AUC	Gain [95% CI]
PhoBERT	0.960	–	0.951	–
+ chord	0.976	+0.015 [0.010, 0.022]	0.970	+0.019 [0.011, 0.026]
+ tone	0.961	+0.000 [-0.001, 0.002]	0.952	-0.001 [-0.003, 0.002]
+ poetic	0.962	+0.001 [0.000, 0.003]	0.953	+0.001 [0.000, 0.003]
+ alignment	0.961	+0.001 [-0.001, 0.002]	0.952	+0.000 [-0.001, 0.002]
+ chord + tone	0.976	+0.015 [0.009, 0.022]	0.970	+0.018 [0.009, 0.025]
+ all features	0.976	+0.016 [0.009, 0.023]	0.970	+0.018 [0.009, 0.025]

Table S6: Fold composition (seed 1): songs / bolero share / groups.

Scheme	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Composer	117 / 45% / 73	115 / 45% / 74	116 / 46% / 71	120 / 46% / 71	115 / 45% / 73
Strict	175 / 64% / 31	101 / 38% / 57	101 / 38% / 58	104 / 38% / 60	102 / 37% / 60

S5. Audio event alignment: onset-window sensitivity

Syllables with a valid pitch track (bolero 4,619, V-pop 11,460). Audio chord change: detector label changes within the window of the syllable onset. Δ: per-song level-tone difference at audio changes; genre difference with 95% song-bootstrap CI and a permutation test of genre labels across songs (10,000 permutations).

Table S7: Sensitivity to the onset window.

Window	Bolero OR	V-pop OR	Inter. OR (p)	Δ diff [95% CI]	Perm. p
50 ms	1.90 [1.24, 2.90]	1.09 [0.87, 1.38]	1.81 (0.019)	0.104 [0.004, 0.199]	0.059
100 ms	1.79 [1.25, 2.56]	1.12 [0.94, 1.32]	1.65 (0.019)	0.107 [0.034, 0.178]	0.003
150 ms	1.53 [1.14, 2.06]	1.17 [1.01, 1.34]	1.35 (0.091)	0.083 [0.017, 0.146]	0.007
200 ms	1.46 [1.16, 1.85]	1.15 [1.01, 1.30]	1.31 (0.062)	0.079 [0.025, 0.130]	0.004
300 ms	1.41 [1.13, 1.76]	1.06 [0.92, 1.21]	1.37 (0.025)	0.108 [0.056, 0.158]	< 0.001

Leave-one-song-out (bolero): OR range 1.45–1.65 with audio changes (150 ms) and 2.35–2.98 with sheet changes.

S6. Vocal–chord compatibility against empirical nulls

Null (a): chord track circularly shifted by a random offset of at least 10 s; null (b): pitch frames of each syllable paired with chord frames of a random other syllable of the same song. Sign test over songs.

Table S8: Song-mean compatibility C vs. nulls.

Variant	Genre	Observed	Null	Songs above null (p)
All frames, null (a)	Bolero	0.463	0.387	27/29 (7.1×10^{-8})
	V-pop	0.437	0.357	51/59 (1.4×10^{-8})
All frames, null (b)	Bolero	0.463	0.382	29/29 (3.7×10^{-9})
	V-pop	0.437	0.366	50/59 (3.7×10^{-9})
Central half, null (a)	Bolero	0.464	0.387	26/29 (2.0×10^{-7})
	V-pop	0.439	0.357	51/59 (1.9×10^{-8})
Median pitch, null (a)	Bolero	0.488	0.387	27/29 (5.2×10^{-8})
	V-pop	0.454	0.357	50/59 (7.1×10^{-9})
Periodicity-weighted, null (a)	Bolero	0.467	0.387	27/29 (5.2×10^{-8})
	V-pop	0.442	0.357	52/59 (8.8×10^{-9})

Table S9: Level-vs-sharp compatibility at events. Cells: mean C for level / sharp syllables; DiD: bolero gap minus V-pop gap with 95% song-bootstrap CI. "Excess" subtracts each syllable's null (a).

Measure	Event	Bolero L / S	V-pop L / S	DiD [95% CI]
Raw C	Chord change	0.521 / 0.378	0.436 / 0.370	+0.077 [0.002, 0.153]
	Line end	0.611 / 0.379	0.534 / 0.436	+0.134 [0.031, 0.236]
Central half	Chord change	0.532 / 0.379	0.439 / 0.376	+0.090 [0.009, 0.174]
	Line end	0.631 / 0.364	0.546 / 0.440	+0.161 [0.040, 0.272]
Median pitch	Chord change	0.565 / 0.365	0.461 / 0.378	+0.118 [0.011, 0.227]
	Line end	0.700 / 0.326	0.581 / 0.447	+0.241 [0.104, 0.373]
Weighted	Chord change	0.527 / 0.379	0.440 / 0.373	+0.081 [0.001, 0.163]
	Line end	0.619 / 0.382	0.540 / 0.438	+0.136 [0.028, 0.233]
Excess over null	Chord change	0.112 / 0.018	0.073 / 0.025	+0.046 [-0.053, 0.147]
	Line end	0.158 / -0.001	0.153 / 0.070	+0.077 [-0.044, 0.202]

Per-genre level-minus-sharp excess (95% CI): bolero at changes +0.094 [-0.001, 0.184], at line ends +0.159 [0.062, 0.271]; V-pop at changes +0.048 [0.006, 0.087], at line ends +0.083 [0.011, 0.146]. GEE three-way terms on excess: level × change × bolero +0.045 [-0.060, 0.150], $p = 0.40$; level × line-final × bolero +0.064 [-0.054, 0.182], $p = 0.29$.